Topic to discuss

→ Runge-Kutta Method of Second order

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Runge-Kutta Method of Second Order (RK2)

It is a numerical method used to solve first-order differential equation.

The formula is:

$$K_1 = h f(x_n, y_n)$$

$$K_2 = h f(x_n + h, y_n + K_1)$$

$$y_{n+1} = y_n + \frac{1}{2}(K_1 + K_2)$$

Here, K_1 = Slope contribution at starting point

K_2 = Slope contribution at predicted next point

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Note:

RK2 method and euler's modified method are connected, but they are not always taught in the exact same way.

In modified euler method, many books use a predictor-corrector format, where first we predict the value of y, the correct it one or more times like we already in last video.

But in RK2, we directly use two slope formula.

K_1, K_2, y_{n+1}

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Numerical Problem:

Q: Solve using RK2 Method, $\frac{dy}{dx} = x+y$

$y(0) = 1$, find $y(0.1)$ take step size $h = 0.1$

Solution: Given, $f(x,y) = x+y$

$x_0 = 0$, $y_0 = 1$ & $h = 0.1$

$\therefore x_1 = x_0 + h = 0 + 0.1 = 0.1$

RK2 formula,

$K_1 = hf(x_n, y_n)$

$K_2 = hf(x_n + h, y_n + K_1)$

$y_{n+1} = y_n + \frac{1}{2}(K_1 + K_2)$

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We need to find : $y_1 = y(x_1) = y(0.1)$, $n=0$

$$K_1 = hf(x_0, y_0)$$
$$= 0.1\,(x_0 + y_0)$$
$$= 0.1\,(0+1)$$

$\therefore K_1 = 0.1$

Now, calculating for K_2

$$K_2 = hf(x_0+h\,,\,y_0+K_1)$$
$$= 0.1\,[x_0+h+y_0+K_1]$$
$$= 0.1\,[0+0.1+1+0.1]$$

$\therefore K_2 = 0.12$

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$$y_1 = y_0 + \frac{1}{2}(K_1 + K_2)$$

$$= 1 + \frac{1}{2}(0.1 + 0.12)$$

$$\therefore y_1 = 1.11$$ Ans.

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Homework Problem.

Q: Solve using RK2 Method, $\frac{dy}{dx} = x^2 + y$

Given $y(0) = 1$, $h = 0.1$

find $y(0.2)$.

Solution: Given,

$$f(x, y) = x^2 + y$$

$$x_0 = 0 \quad , \quad y_0 = 1 \quad , \quad h = 0.1$$

RK2 formula is,

$$K_1 = h f(x_n, y_n)$$

$$K_2 = h f(x_n + h, y_n + K_1)$$

$$y_{n+1} = y_n + \frac{1}{2}(K_1 + K_2)$$

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first Approximation :

$x_0 = 0$, $y_0 = 1$, $h = 0.1$, $n = 0$

$x_1 = x_0 + h = 0.1$

Calculating K_1 ,

$$K_1 = h f(x_0, y_0)$$

$$= 0.1 \left[x_0^2 + y_0 \right]$$

$$= 0.1 (0 + 1)$$

$$\therefore K_1 = 0.1$$

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Calculating K_2,

$$K_2 = hf(x_0+h,\ y_0+K_1)$$

$$= 0.1\left[(x_0+h)^2 + (y_0+K_1)\right]$$

$$= 0.1\left[(0+0.1)^2 + (1+0.1)\right]$$

$$\therefore K_2 = 0.111$$

Calculating y_1,

$$y_1 = y_0 + \frac{1}{2}(K_1+K_2)$$

$$= 1 + \frac{1}{2}(0.1+0.111)$$

$$\therefore y_1 = 1.1055$$

So, $y(0.1) = 1.1055$

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Second Approximation :

$x_1 = 0.1$, $y_1 = 1.1055$, $n = 1$, $h = 0.1$

Calculating K_1 ,

$$K_1 = h f(x_1, y_1)$$

$$= 0.1\,(x_1^2 + y_1)$$

$$\Rightarrow K_1 = 0.1\left[0.1^2 + 1.1055\right]$$

$$\therefore K_1 = 0.11155$$

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Calculating K_2,

$$K_2 = h f(x_1 + h, y_1 + K_1)$$

$$= 0.1 \left[(x_1 + h)^2 + (y_1 + K_1) \right]$$

$$= 0.1 \left[(0.1 + 0.1)^2 + (1.1055 + 0.11155) \right]$$

$$\therefore K_2 = 0.125705$$

Calculating y_2,

$$y_2 = y_1 + \frac{1}{2}(K_1 + K_2)$$

$$= 1.1055 + \frac{1}{2}(0.11155 + 0.125705)$$

$$\therefore y_2 = 1.224127$$

So, $y_2 = y(x_2) = y(0.2) = 1.224127$ Ans.

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